Data Analytics

A Journey Through Loss: Tracing Looted Yemeni Artefacts

From Excavation to Circulation: A Data-Driven View

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Introduction

Business Use Case / Goal / Plan

One of the many areas of archaeological research focuses on the study of destroyed and looted sites during conflicts. While the field of archaeology is increasingly making use of digital tools and technologies, it has historically lagged other domains. The same can be observed in the heritage sector (artworks, museums), where many institutions do not even have exhaustive inventories (personal discussion with ICOM[1] experts), and sometimes not even digital ones. Archaeological pieces are often looted directly from sites or from museums where they were stored. They then circulate across the world, sometimes reappearing on auction platforms or even in museum collections.

Several initiatives aim to identify and track dispersed objects (whether missing or for sale) to fight illicit trafficking, which often finances armed groups and, more broadly, can be linked to ideological attempts to destroy cultural heritage. In January, we attended the ANCHISE forum[2] in Athens with a former colleague, which specifically addresses data and the protection of cultural heritage objects. This colleague also shared her intention to pursue a PhD on looted objects in Yemen. To her surprise, many experts were not aware of this existing catalog and showed strong interest in accessing her work. In this context, I wanted to explore how data from the relevant sources could be extracted and structured into an organized, accessible, and queryable format.

This initial objective took the form of a proof of concept (POC), aiming to demonstrate that a complete data pipeline could be built. It was then extended by transforming object descriptions and images into vector data, allowing similarity comparisons between looted objects and those appearing on online art market platforms.

Finally, structuring this data made it possible, in the case of Yemen, to outline one or several profiles of at-risk objects (type, material, trajectories), as well as patterns of circulation and destinations that can contribute to and further inform research on illicit trafficking.

Project Planning

<u>The project was divided into two main phases:</u>

The first phase (8 days) involved selecting the project (from 3 initial personal proposals), defining the POC[3] and MVP[4], which allowed us to prioritize and organize activities in order to demonstrate the project's feasibility, scale it up, and meet the specifications of the final project, concluding the bootcamp and resulting in an initial release on GitHub.

The second phase (7 days) involved improving the code (reviewing and cleaning up the code, due to extensive research and testing), completing new data sources, the RNCP presentation and report, and exposing the data via an API as well as pushing it to Cloud SQL and BigQuery, in order to meet the required specifications.

The task dashboard was created on Trello

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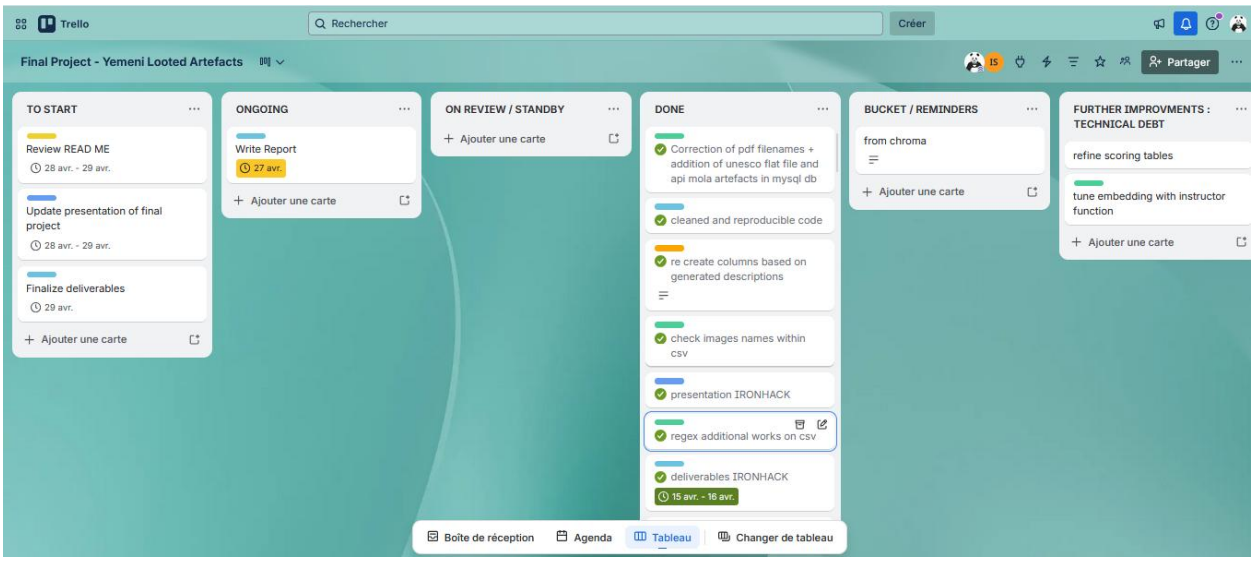
[1] https://icom.museum/fr/ international council of museums

[2] https://www.anchise.eu/finalforum

[3] Proof of Concept

[4] Minimum Valuable Product



The board follows a classic Kanban-style project management approach for software development

Lists:

- **To Start**: Tasks to be started
- **On Going**: Tasks currently in progress
- **Review / On Standby**: Tasks that are complete but require review, or are on standby either because they need to be reworked or refined, or because they have been deprioritized (e.g., if errors need to be corrected)
- **Done**: Tasks fully completed
- **Bucket / Reminders**: Additional information (links to data sources, text to keep, tips)
- **Further improvements/technical debt**: Technical debt or major areas for improvement, particularly in the future. For example, in our case, adding an instructor during data vectorization in Chroma.

Each card has at least <u>a due date and a label</u>. The labels correspond to colors used to classify cards by theme (green = code)

Cards at the top of the list are prior.

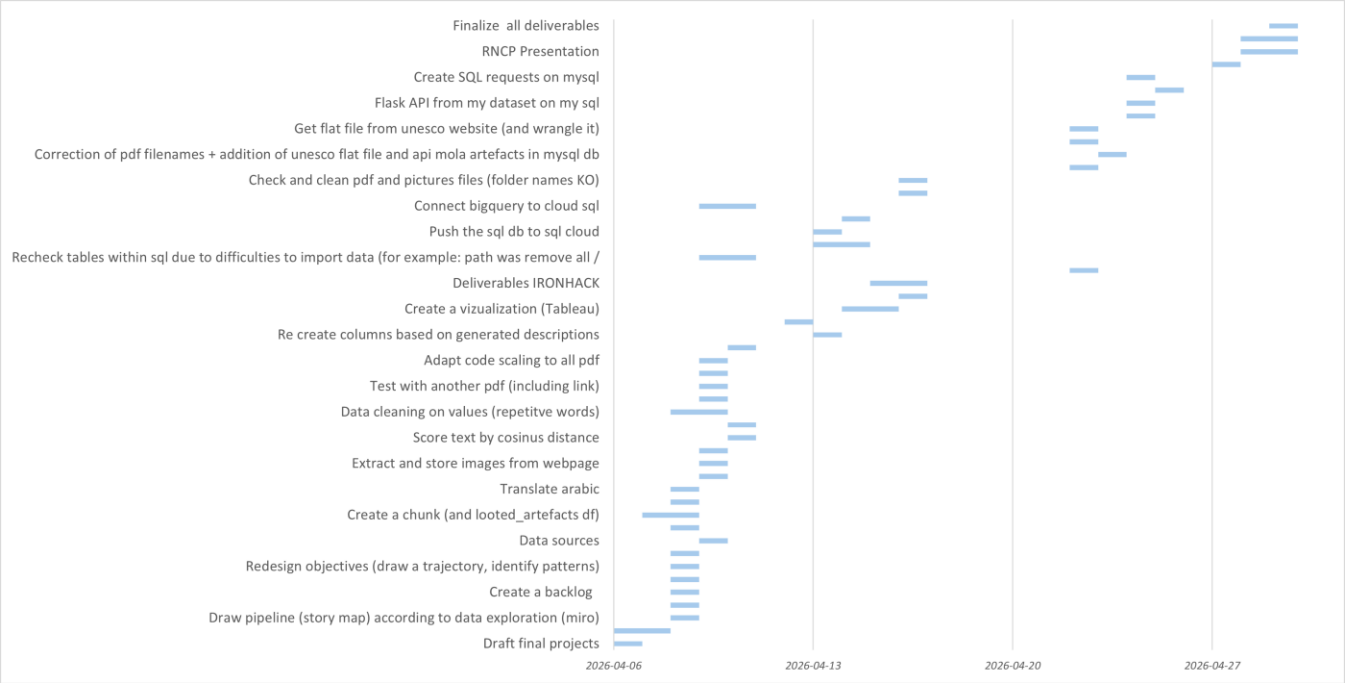
Lean backlog (since only one person is working on the project) and labels key metrics

Label	% of total time
Code	56%
Project design	19%
Infra (SQL)	15%
Demo / Viz	8%
Multiple	2%

As expected, the bulk of the project was devoted to scripting, particularly data cleaning and transformation. A fair amount of time was spent planning the organization of tasks to ensure efficiency within the scope of the POC.

Task	Start Date	Duration	Label
Draft final projects	2026-04-06	1	code
Data exploration and brain sotrming	2026-04-06	2	project design
Draw pipeline (story map) according to data exploration (miro)	2026-04-08	1	project design
Prioritize tasks to be efficient (POC) risk + added value = > MVP	2026-04-08	1	project design
Create a backlog	2026-04-08	1	project design
Set up the environment and repo	2026-04-08	1	infra/deliv
Redesign objectives (draw a trajectory, identify patterns)	2026-04-08	1	project design
Watching a talk about vectorial database to understand what it is	2026-04-08	1	project design
Data sources	2026-04-09	1	project design
Define a consistent planning	2026-04-08	1	project design
Create a chunk (and looted_artefacts df)	2026-04-07	2	code
Extract and store images	2026-04-08	1	code
Translate arabic	2026-04-08	1	code
Extract data from webpage (create art_artefacts)	2026-04-09	1	code
Extract and store images from webpage	2026-04-09	1	code
Create vectorial db (text, images)	2026-04-09	1	code
Score text by cosinus distance	2026-04-10	1	code
Score by comparing datasets	2026-04-10	1	code
Data cleaning on values (repetitve words)	2026-04-08	2	code
Clean key_en (all pdfs within the df)	2026-04-09	1	code
Test with another pdf (including link)	2026-04-09	1	code
Webscrap	2026-04-09	1	code
Adapt code scaling to all pdf	2026-04-09	1	code
Create relational db (ERD + mysql)	2026-04-10	1	infra/deliv
Re create columns based on generated descriptions	2026-04-13	1	code
Locating city, country and type of salers	2026-04-12	1	code
Create a vizualization (Tableau)	2026-04-14	2	viz
Presentation IRONHACK	2026-04-16	1	demo
Deliverables IRONHACK	2026-04-15	2	infra/deliv
Code review and cleaning	2026-04-22	1	infra/deliv
Recheck tables within sql due to difficulties to import data (for exampl	2026-04-09	2	code
Numerous issues with csv and characters, languages (despite of rege	2026-04-13	2	code
Push the sql db to sql cloud	2026-04-13	1	code
Regex additional works on csv	2026-04-14	1	code
Connect bigquery to cloud sql	2026-04-09	2	infra/deliv
Update web images names	2026-04-16	1	code
Check and clean pdf and pictures files (folder names KO)	2026-04-16	1	code
Check images names within csv	2026-04-22	1	code
Correction of pdf filenames + addition of unesco flat file and api mola	2026-04-23	1	code
Get data from API mola omeka	2026-04-22	1	code
Get flat file from unesco website (and wrangle it)	2026-04-22	1	code
Review looted artefacts map (number of looted artefacts instead ofnu	2026-04-24	1	code
Flask API from my dataset on my sql	2026-04-24	1	code
Denormalized on Big Query	2026-04-25	1	infra/deliv, code, viz
Create SQL requests on mysql	2026-04-24	1	code
Write RNCP Report	2026-04-27	1	infra/deliv
RNCP Presentation	2026-04-28	2	demo
Update READ ME	2026-04-28	2	project design
Finalize all deliverables	2026-04-29	1	infra/deliv

Gantt diagram of the full project



Data and data sources

Web scraping – Looted artefacts catalogue from Yemen (https://goam.gov.ye/Looted): this is the dataset I wanted to explore in depth. It responds to a real need (a friend applying for a PhD on this topic asked whether it would be possible to build a database), while also aligning with the skills developed during the bootcamp (API, web scraping, data analysis). It is also a subject that matters to me personally, as I am involved in local associations promoting environmental and cultural heritage. More broadly, I am aware of the effort made by those documenting what has been taken from them — this goes far beyond financial loss. The first cataloguing efforts started only one month after the beginning of the war, and many studies highlight a form of cultural looting driven by ideological destruction.
Concretely, this involved web scraping to download PDFs corresponding to inventories of looted artefacts. Some artefacts were also linked to external websites (art dealers or commercial platforms), which led to a second round of web scraping. Descriptions and images from both sources were then vectorized to compare them through similarity analysis (cosine distance).

API - MOLA artifacts (https://mola.omeka.net/)
The Museum of Looted Antiquities aims to document stolen artefacts that have been returned to their country of origin. Their API was used to build a table of repatriated objects. This dataset made it possible to identify origin and return countries, and to compare them with the trajectories of Yemeni artifacts to explore potential patterns.

Flat file - UNESCO World Cultural Heritage (https://whc.unesco.org/en/list/)
UNESCO lists cultural and natural heritage sites and identifies those considered at risk. The data is available as an Excel file. After cleaning and querying it (SQL), I used it to assess whether Yemen is among high-risk countries. This helped confirm the relevance of focusing on this case.

Databases: All these sources were integrated into a single relational database (sql/dump/new_yemen_looted_artefacts.sql), which was then deployed on Cloud SQL.
A vectorial database has been experimented with Chroma DB in a sqlite format for text and images embeddings and comparisons.

BigQuery: BigQuery was used on top of Cloud SQL to create a denormalized version of the dataset (sql/bigquery_denorm_query.sql), in line with reporting requirements. Data on looted antiquities is rarely fully accessible in open source: in most cases, only partial access is available (as with MOLA), or specific authorization is required. Many databases in this domain are sensitive (e.g., Interpol), which highlights the importance of structuring and consolidating the available data.

Metadata/Data dictionary

The tables contained in the MySQL database are presented here (it can also be viewed in CSV format in the /data/clean directory).
Note: For better readability, data types and lengths are not listed below. They can be found either in the ERD (/figures) or by uploading the SQL dump (/sql/dump).

art_dealers (# 64): web platforms where Yemeni looted artifacts have been related to	
main_website	Original domain within provided url from pdf
normalized_domain	Cleaned domain of the art dealer
city	City
country	Country
source_classification	Categorization of art dealers (/slides. source_classification_metadata.xlsx)

looted_artefacts (# 1,125) (last update February 2026): structured table of looted Yemeni artifacts from PDF catalogues	
artifact_id	Unique ID based on pdf name and page number
pdf_name	Pdf filename
page_num	Page number
type	Main type of the artifact (currency, sculpture, jewelry…)
detail	Sub-type (coin, statute, ring)
material	Materials of the artifact (iron, gold, marble)
iconography	Visual image / symbol (Human, animal..)
period_general	Post-medieval / Modern Islamic Medieval Late Antiquity South Arabian Antiquity Unclassified
period_precise (time intervals are related to perdio general, following the same order)	18 – 20th 8 – 10th 4 – 6th 1 – 3rd Unknown

Pdf_images (# 985): images extracted from the pdf files	
artifact_id	ID based on pdf name and page number
image_name	Unique ID based on pdf name, page number and image number
image_path	Folder of the image

Web_pages (# 685): images extracted from the webpages	
article_id	Unique ID of the web page
artifact_id	ID based on pdf name and page number
link	urlL of the artifact page
text_for_matching_web	Concatenated text descriptions from the page
price	Price of the artifact (if present)

Web_pages (# 685): images extracted from the webpages	
currency	Currency (if present)
wb_txt_shrt	Restructured description in short format
wb_txt_lg	Restructured description in full format
main_website	Home url

Web_photos (# 195): extracted images of the art dealers urls	
artifact_id	ID based on pdf name and page number
image_name	Unique
image_path	Folder of the image
article_id	ID of the web page

Match_scoring (# 246): similarity scoring of descriptions and images of artifacts (pdf versus web)	
index	Unique ID
artifact_id	ID based on pdf name and page number
score_img	Best similarity (distance) score between images (pdf / web) of the looted artifact
score_txt	Similarity (distance) score between restructured descriptions (pdf / web) of the looted artifact
score_type	score_txt_only, score_img only, combinated_score
final_score	If combinated, mean of score_img and score_txt

Mola_artefacts #167 (documented version of data 2024): returned looted artefacts with identified countries	
id	Unique ID auto-incremented
object_type	Main type of the artifact (currency, sculpture, jewelry…)
object_materials	Materials of the artifact (iron, gold, marble)
sending_country	Country returning the artifact
receiving_country	Country receiving the artifact (usually country of origin)
year_returned	Year of return
year_acquired	Year of acquisition
date_looted	Year of looting (sometimes estimated)

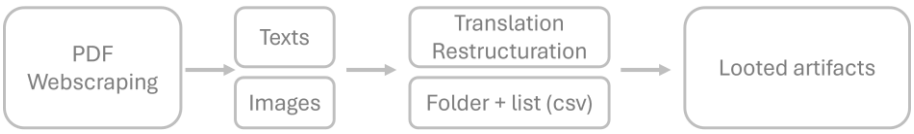
Unesco_sites_in_danger #167 (las update summer 2025): Countries with 1 or more cultural/environmental UNESCO site(s), characterised by their status (danger/no danger)	
states_name_en	Country name (in English)
no_danger	Number of UNESCO sites in the country not listed as endangered
yes_danger	Number of UNESCO sites in the country listed as endangered
percent_danger	Percentage of sites in the country that are endangered

Data collection

Pipeline schemes and main steps

The proof of concept (POC) aimed to 1) build a data pipeline from pdf extraction to API exposure 2) compare text and images per artifact by embedding data. In this chapter, we propose to represent data pipeline, main steps and outputs through simplified schemes. Related scripts are indicated and can be found in /notebooks, /python_files and /sql.

Process 1: Get a table and extracted images from downloaded pdf files of Yemeni looted artifacts



Extracting and translating information and images per artifact from various pdf frames (*pdf_webscraping.ipynb, pdf_chunk_extraction.ipynb, data_wrangling_cleaning.ipynb*)

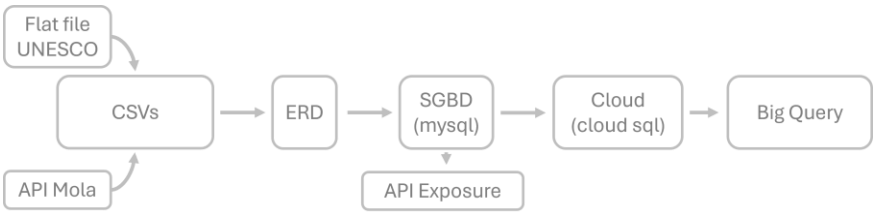
Process 2: Get a table and extracted images from web pages related to the looted artifacts

Extract relevant data with minimum noise (*art_dealer_image_extraction.py, art_dealer_text_extraction.py*)

Process 3: Comparing text and images from both sources to assess similarity of artifacts

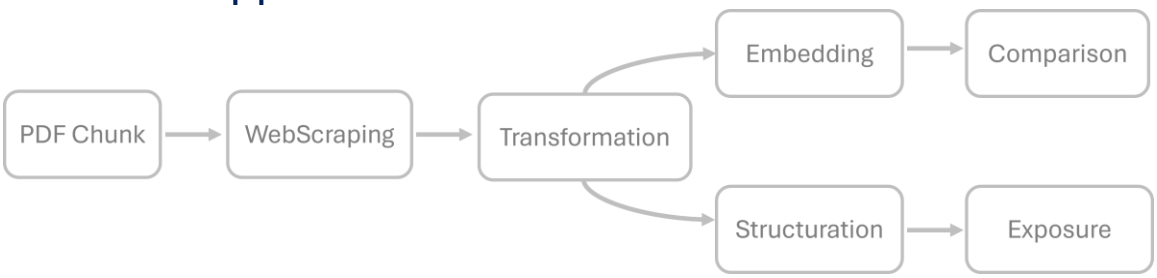
Analyzing how similar descriptions and images are to demonstrate data vectorization is a new relevant asset in illicit trafficking analysis (*vector_db_comparison.ipynb*)

Process 4: Add complementary data. Structure in a relational database and expose it through an API



Build a database, query and share *(mola_api_data.ipynb, flask_api.py, /sql)*

End to end data pipeline



The scripts must be run in the following order

Pipeline from pdf extraction to vector comparison

pdf_webscraping.ipynb
Download PDFs from main source

pdf_chunk_extraction.ipynb:
Create 2 csv files (artifacts and artifacts_images) and 1 images' folder

data_wrangling_cleaning.ipynb part 1
data cleaning and creating 1 csv "web_links" for second webscraping

art_dealer_image_extraction.py
Extract images from art dealers' pages, create 1 folder and 1 csv file

art_dealer_text_extraction.py
Extract text from art dealers' pages, create 1 csv file

data_wrangling_cleaning.ipynb part 2
Data cleaning and creating a "parent" csv including all structured and translated information from both sources (pdf and web), including data cleaning on pictrures folders, lists. create 1 csv for art dealers. finalize data for erd and sql import

vector_db_comparison.ipynb
Data embedding, vectors collections comparison based on cosine distance queries, create csv match_scoring csv + EDA

Scripts that can be run independently but before API exposure and sql import

mola_api_data.ipynb
Get data from mola api + EDA

unesco_sites.ipynb
Reshape data from unesco flat file + EDA

Post db creation

flask_api.py
Expose mysql via API Flask

Data cleaning and wrangling

Data Cleaning / Wrangling

Data cleaning and wrangling were processed during the entire lifetime of the project. Therefore, corrections were applied with both Python and SQL scripts.

- General text cleaning using a lot of **regexes** (Cleaning of encodings, special characters, and extra spaces, filenames)
- Removing/ replacing null values
- Replacing column names
- Creating specific **functions** (i.e remove duplicated words)
- Extraction and standardization of **web links** (urlparse)
- Deletion of duplicated images (images based **on hash5**)
- **Translation from Arabic to English**: prior experiments using MarianMTModel and MarianTokenizer (Helsinki-NLP, Hugging Face) did not yield satisfactory results. Exploring open-source solutions remains an essential step in academic research, and the code has been retained to document this process before selecting a paid model.
- Normalization of descriptive fields
- Standardization of centuries and historical periods

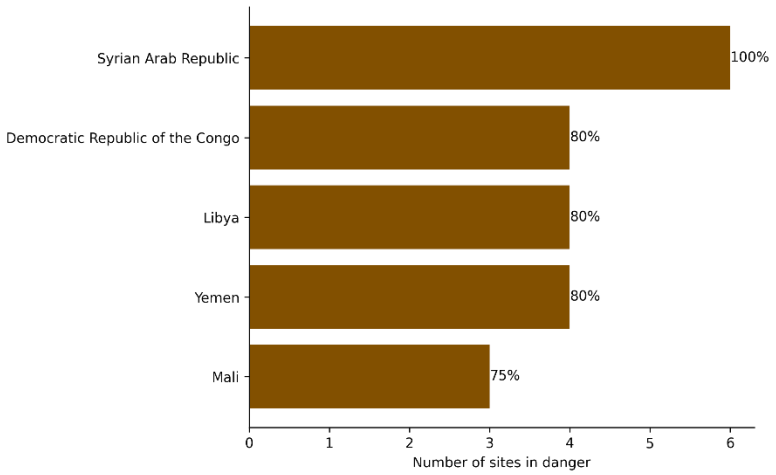
Note: additional data wrangling (and querying) will be developed in the SQL scripts section

```python
# define a common function for all regex issues
import unicodedata
from ftfy import fix_text

def clean_text_and_pray(x):
    if pd.isna(x):
        return ""
    x = fix_text(x) #clean weird chars
    x = str(x) #to add before try
    x = x.replace("Ã–", "×")
    x = x.replace("â€“", "–")
    x = x.replace("â€”", "—")
    x = x.replace("â€™", "’")
    x = x.replace("â€˜", "‘")
    x = x.replace("â€œ", "“")
    x = x.replace("â€\x9d", "”")
    x = x.replace("Â ", " ")
    x = x.replace("Â", "")
    try:
       x = x.encode("latin1").decode("utf-8")
    except (UnicodeEncodeError, UnicodeDecodeError):
        pass
    x = unicodedata.normalize("NFKC",x) # normalization unicode
    x = x.replace("\n"," ").replace("\r"," ").replace("\t"," ") # tabs and backlines
    x = x.replace("\u00a0"," ") # - when a text is cut non breaking space
    x = x.replace(";"," ") # - when a text is cut non breaking space
    x = re.sub(r'[\u0600-\u06FF\u0750-\u077F\u08A0-\u08FF\uFB50-\uFDFF\uFE70-\uFEFF]+','',x)
    x = re.sub(r"\s+"," ",x).strip()  #normalize to keep only 1 space no use for path might h
    x = re.sub(r'[\x00-\x1F\x7F]', '',x)  # control ASCII
    x = re.sub(r'[\u200B-\u200D\uFEFF]', '',x)  # invisible
    x = re.sub(r"0-[_\s]+", "0-", x)
    return x
```

```python
web_images_folder = "../figures/web_photos/"

# get an identifier
def hash_file(path):
    #get the unique hash MD5 of the image
    h = hashlib.md5()
    with open(path, "rb") as f:
        h.update(f.read())
    return h.hexdigest()

# main function
def remove_duplicate_images(folder):
    seen_hashes = set()
    removed_images = []

    for filename in os.listdir(folder):
        img_path = os.path.join(folder, filename)

        # ignore subfolders
        if not os.path.isfile(img_path):
            continue

        try:
            h = hash_file(img_path)
        except PermissionError:
            print(f"Permission refused for {img_path}, skipping ...")
            continue

        if h in seen_hashes:
            os.remove(img_path)
            removed_images.append(filename)
        else:
            seen_hashes.add(h)
```

Data Exploratory and Visualization

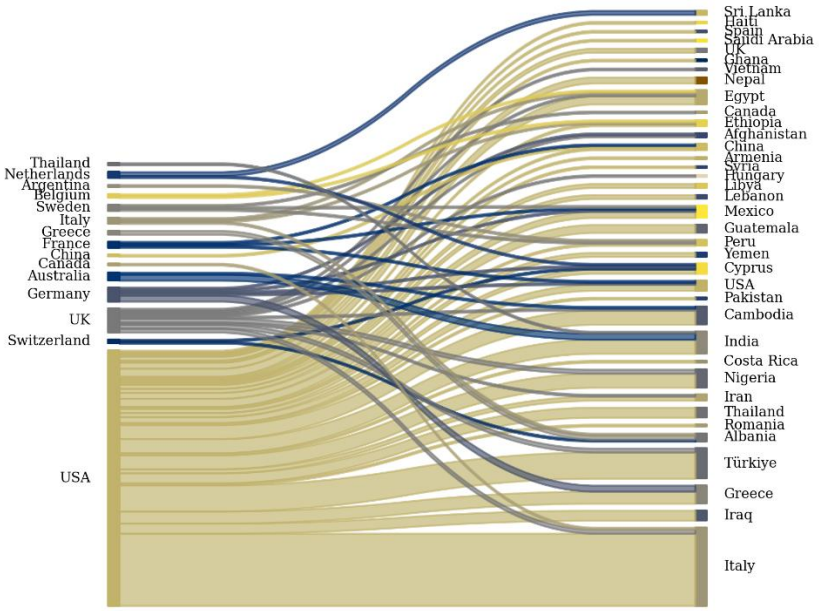
UNESCO World Heritage Sites: By summing the number of sites at risk and those not at risk per country, we have created an initial bar chart that identifies the countries most at risk. This chart is based both on the absolute number of sites at risk and on their proportion of the total number of sites in each country. It thus highlights the countries with the highest concentration of threatened sites and where these sites account for most of all sites registered.



It is noteworthy that Yemen is one of five countries that have both a high number of sites in danger (4) and account for most of its UNESCO-listed heritage (80%).

It shares second place with the DRC and Libya, behind Syria. This finding confirms the reality of the looting that has been documented.

Mola artifacts: We retained only the artifacts for which both the country of origin and the country of return were specified to build a Sankey diagram. In addition, we calculated (using SQL) the average time taken by each country to return looted objects, while also tracking the total number of artifacts returned by each country.

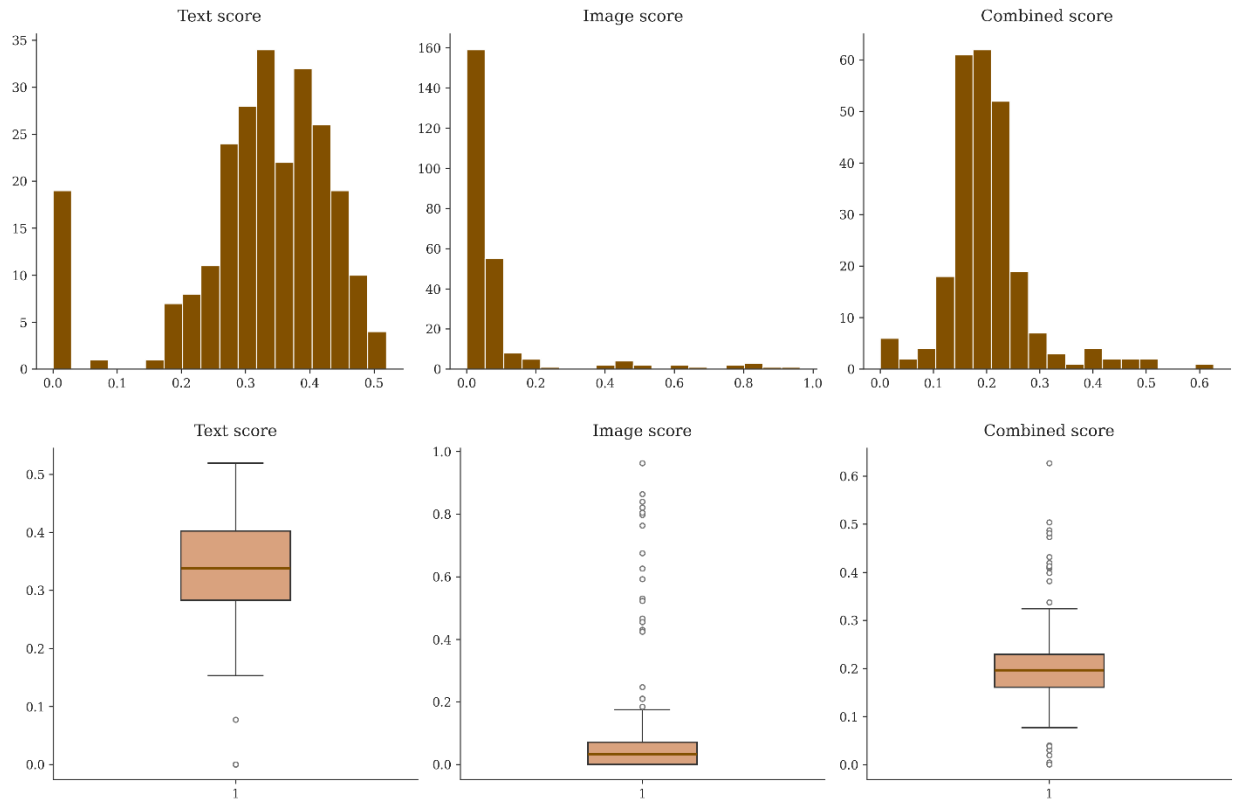


USA, UK, the Netherlands, and other European countries are well represented in the returns of artifacts. These results remain limited due to a non-exhaustive and relatively recent database, but they do provide insight into the general patterns of how these objects circulate. The average return duration per country indicates a long process before getting artifacts back.

sending_ country	count (*)	avg_duration (years)
Netherlanc	2	96
Sweden	2	78
Italy	1	68
Germany	2	39
USA	74	24
Belgium	2	18
Thailand	1	18
Australia	3	10
UK	2	6
France	1	0

Comparing images and texts from PDF and webpages:

The cosine distance is a measure used to assess the similarity between two vectors by comparing their orientations. It is derived from the cosine similarity, which calculates the cosine of the angle formed between two vectors: the closer the value is to 0, the more similar the vectors are. The cosine distance is generally defined as 1 - cosine similarity, meaning that a value close to 0 indicates a high degree of similarity, based on semantic or visual proximity. In other words, a value of 0,3 indicates 70% of similarity.
Only images or textual descriptions belonging to the same artifact were compared. This process relies on the artifact_id stored in tables and the vectorization of descriptions and images, which we explain in detail in the following section (databases).
Descriptions of a given artifact from pdf and web were compared, providing a score for text. Pictures were also compared but as an artifact could gather many pictures from pdf and web, only the best scoring between 2 images was kept, providing an image score. When an artifact combines text and description scores, then the final score for this artifact corresponds to the mean of both.



Overall, the matching scores are very high, with low means of 0,3 for text and 0,08 for images. Most text pairs show moderate similarity (around 70%), with very few outliers. Regarding images, the similarity scores indicate many highly similar pairs (distance close to 0 that suggest they are identical), although a few significant outliers remain (distance above 0,5). In other words, descriptions and images are mostly similar. Interpretation will be detailed in Conclusions' section.

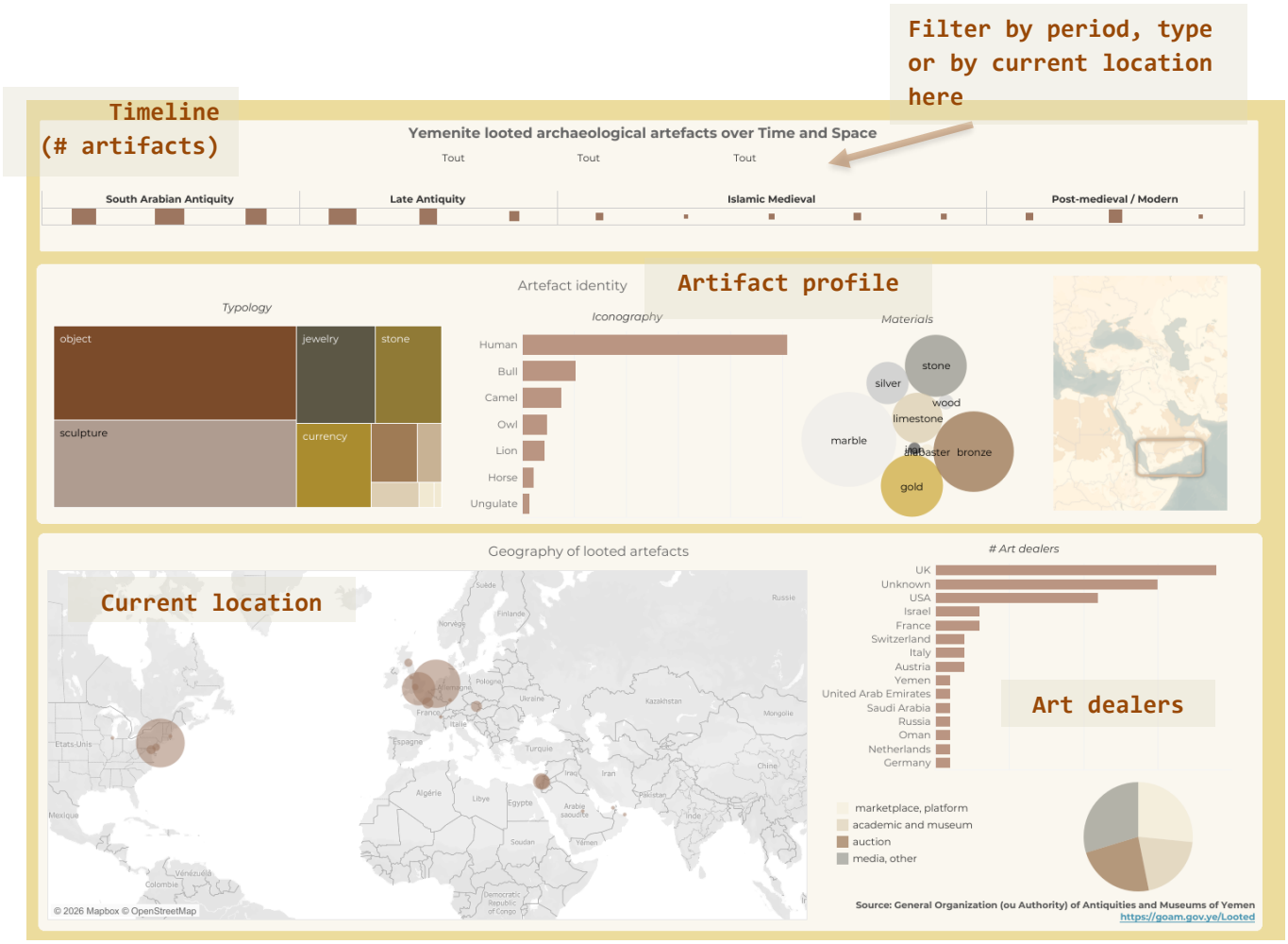
Average duration for returning artifacts

An analysis of the average duration for the return of looted cultural artifacts shows that the United States leads the way in terms of both volume and speed, with 74 cases and an average duration of 24 years, while European countries such as the Netherlands and Sweden have significantly longer average return times (96 and 78 years, respectively). There is evidence of very long delays in the process of returning stolen objects.

sending_country	count(*)	avg_duration
Netherlands	2	96
Sweden	2	78
Italy	1	68
Germany	2	39
USA	74	24
Belgium	2	18
Thailand	1	18
Australia	3	10
UK	2	6
France	1	0

Data visualisation on Tableau

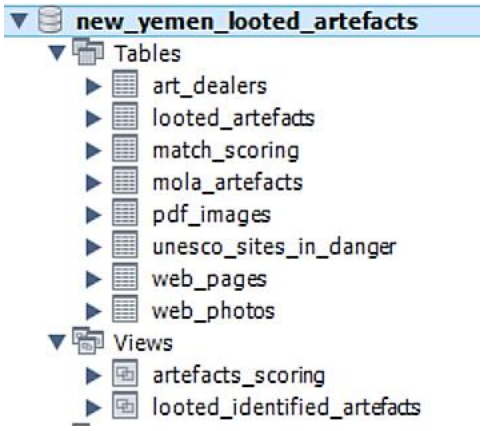
By linking the various tables in the Yemeni Looted Artifacts dataset, we were able to create a dynamic dashboard in Tableau Public. The looted artifact profile and trajectory will be detailed in the conclusions section. In the meantime, you can take a look at the dashboard by clicking on the link above or directly on the image to get your own insights!


Filter by period, type or by current location here
Timeline (# artifacts)
Yemenite looted archaeological artefacts over Time and Space
Tout
Tout
Tout
South Arabian Antiquity
Late Antiquity
Islamic Medieval
Post-medieval / Modern
Artefact identity
Artifact profile
Typology
object
jewelry
stone
sculpture
currency
Iconography
Human
Bull
Camel
Owl
Lion
Horse
Ungulate
Materials
stone
silver
wood
limestone
marble
alabaster
bronze
gold
Geography of looted artefacts
Current location
© 2026 Mapbox © OpenStreetMap
Art dealers
UK
Unknown
USA
Israel
France
Switzerland
Italy
Austria
Yemen
United Arab Emirates
Saudi Arabia
Russia
Oman
Netherlands
Germany
Art dealers
marketplace, platform
academic and museum
auction
media, other
Source: General Organization (ou Authority) of Antiquities and Museums of Yemen
https://goam.gov.ye/Looted

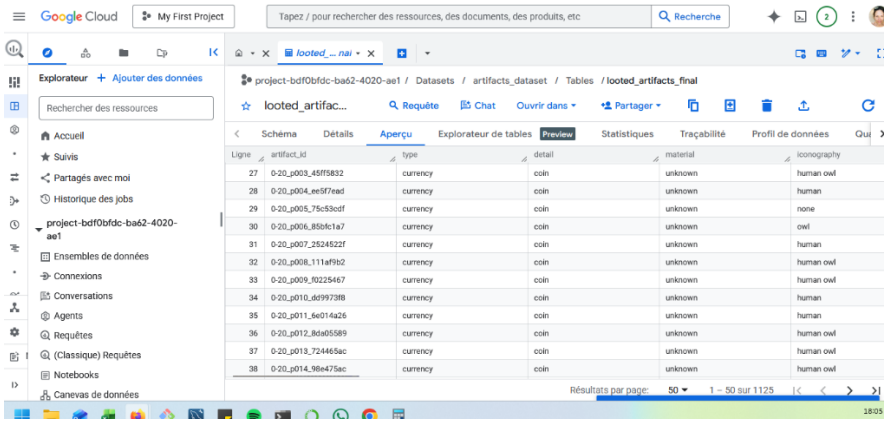
Database type selection

Relational database: from mysql to BigQuery

In this project, the data is split across several connected tables (artefacts, web pages, images, dealers), with shared identifiers linking them together. A relational database therefore fits naturally with this structure. It allows to organize the data in a consistent way, avoid duplicating information (for example the same dealer or artefact appearing multiple times), and maintain coherence between tables through keys and constraints. It also makes it much easier to explore the data, especially combining information from different sources (PDFs, web scraping, images). Using SQL is particularly useful to run more advanced and interoperable queries, such as joins and aggregations, which are central to analyze data.
The database was implemented using MySQL, deployed through Google Cloud SQL for managed storage and scalability.
The full dump can be found in folder *sql/new_yemen_looted_artefacts.sql.*



BigQuery was also used **to create a denormalized table from all tables. Indeed, no open access datasets about looted artifacts are available. The Big Query query is accessible in /sql/***bigquery_denorm_query.sql.*



Vectorial database: Chroma DB

To enable similarity comparison between images, and between descriptions, the data was first vectorized using embeddings. No comparison was made between images AND texts (many to many).

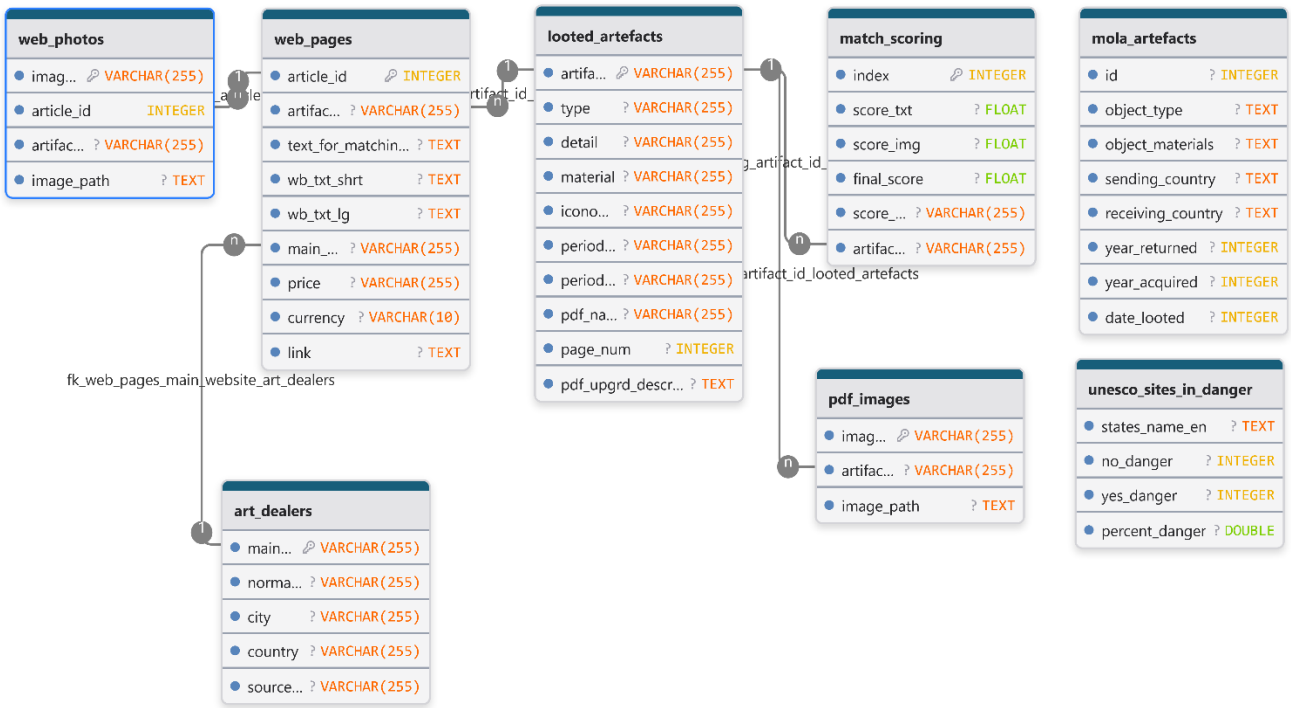
Descriptions were encoded using SentenceTransformers (MiniLM), while images were processed using OpenCLIP and stored as collections in ChromaDB. Sotring embeddings in a vector database wasn't mandatory but relevant to explore. In addition, the syntax is intuitive and easy to familiarize with. The database is in *notebooks/chroma_db/ chroma.sqlite3*

Entities. ERD and SQL Scripts

ERD (Entity Relational Diagram)

was designed in [DrawnDB](), implementing cardinal relationships, primary and foreign keys and fields' types. It turned out to be necessary to switch from the varchar (255) data type, as its length was insufficient to store the descriptions, particularly in text format.

The main tables (without foreign keys) are the looted_artefacts and art_dealers. Mola artifacts and Unesco sites in danger are stored in the DBMS to gather all information in one place but as additional data they are not directly linked to the initial dataset.



SQL Scripts (including Big Query)

We recorded several sql scripts serving different purposes:

1) **Capitalizing a set of swiss-knife scripts** *sql/load, output, helpers.sql*:

In practice, we encountered difficulties importing and exporting data from MySQL due to the relatively difficult CSV parsing caused by the Arabic-to-English translation. As a result, we have scripts and tools for importing/exporting and manipulating data

2) Exploring the data through targeted analyses *sql/SQL_Views_DataInsights.sql*

These analyses rely mainly on SQL aggregation using group by, combined with functions such as count() and avg(), as well as ranking functions (dense_rank) and common table expressions (cte) to structure more advanced calculations. These queries aim to identify the most common artifact types and periods, assess high-risk countries for UNESCO sites, analyze artifact return timelines by country, and highlight the main dealers associated with looted artifacts.

3) Preparing views for the Flask API: *sql/SQL_Views_DataInsights.sql*

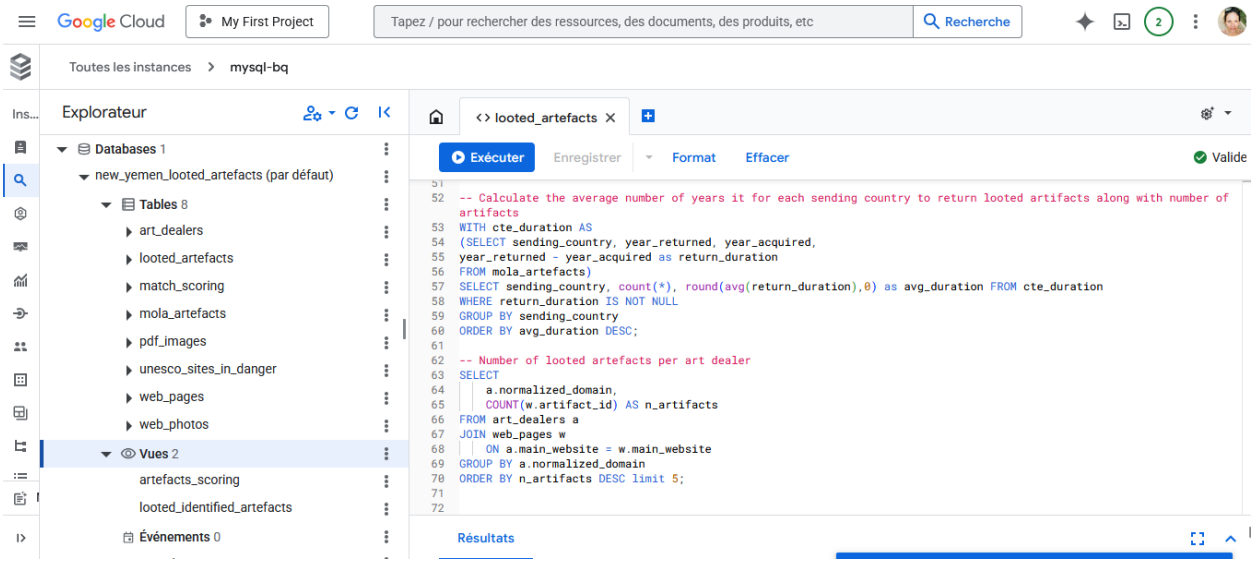
“artefacts_scoring” groups text/image matching scores with PDF descriptions, web texts, and associated links. “looted_identified_artefacts” groups artifacts with their web source, normalized domain, city, country, and source type.

4) Denormalized table in Big Query: *sql/ bigquery_denorm_query.sql*

To finalize the data pipeline, we first migrated the entire database from MySQL to Google Cloud SQL so that we could then query it using BigQuery. We created a table using multiple joins to consolidate all the data from the “looted artifacts,” “webpages,” and “art dealers” tables. Note: the result of the query can be found in *data/clean/ looted_artifacts_final* (XMLSpy)

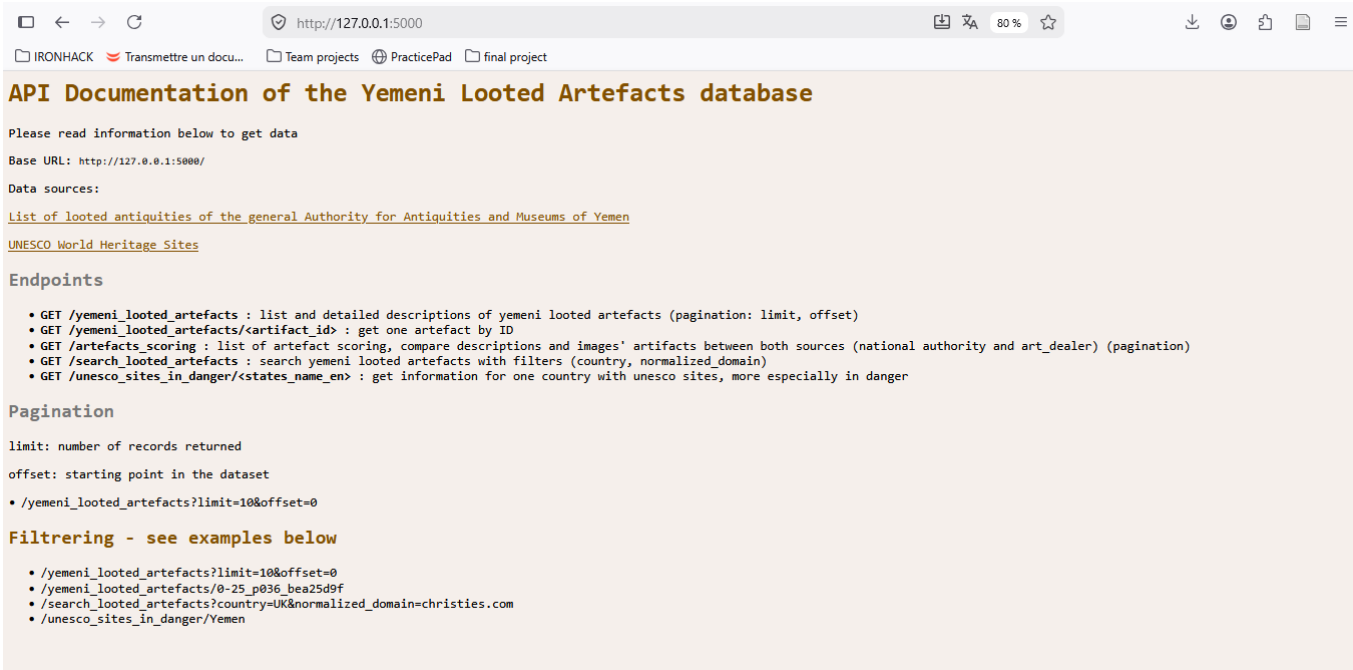
Screenshots below

```sql
-- Assign a specific rank for countries in  high-risk level UNESCO sites in danger based on both volume and percentage (cte)
WITH cte_ranked AS
(SELECT *, DENSE_RANK() OVER (ORDER BY yes_danger DESC) AS Rank_nb,
DENSE_RANK() OVER (ORDER BY percent_danger DESC) AS Rank_pct
FROM unesco_sites_in_danger)
SELECT states_name_en, no_danger, yes_danger, percent_danger,
(Rank_nb * Rank_pct) AS final_rank
FROM cte_ranked;
```



API

Exposing data via an API allows to easily access to structured datasets that can therefore be reused by other applications. It enables real-time querying, improves interoperability between systems, and provides a flexible way to share data. For this project, we used Flask to build an API, creating several endpoints along with a home route that provides **API documentation**, as shown below:



To activate the API, we simply need to run the Python file from the terminal.

```
(final_project_looted_artefacts) (base) C:\Users\Utilisateur\De
sktop\IRONHACK_DA\COURSES\FINAL_PROJECT\final_project_looted_ar
tefacts>python main.py
 * Serving Flask app 'main'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a produ
ction deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 492-818-721
127.0.0.1 - - [27/Apr/2026 23:30:28] "GET / HTTP/1.1" 200 -
```

Main characteristics

We use different endpoints, methods and parameters to answer to required specifications and provide targeted and flexible information (this script has been stored in *python_files/ flask_api.py*).

- Cursor: execute SQL query and return result
- Limit/Offset: paginate results (number of results and starting point in the dataset when using Flask)
- Get: retrieve data
- Search: Get with filters
- fetchone vs fetchall: one result / multiple results

As you can see from the API documentation main page, 5 endpoints were created for different use cases:

1) yemeni_looted_artefacts: list and detailed descriptions of yemeni looted artefacts (pagination: limit, offset)
2) yemeni_looted_artefacts/<artifact_id>: get one artefact by ID
3) artefacts_scoring: list of artefact scoring, compare descriptions and images' artifacts between both sources (national authority and art_dealer) (pagination)
4) search_looted_artefacts: search yemeni looted artefacts with filters (country, normalized_domain)
5) unesco_sites_in_danger/<states_name_en>: get information for one country with unesco sites, more especially in danger

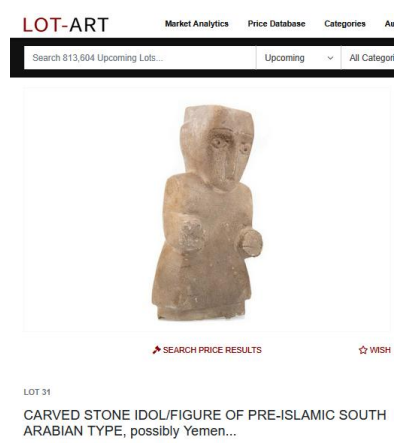
Some examples: combination of API documentation url /endpoint

http://127.0.0.1:5000//yemeni_looted_artefacts?limit=50&offset=20


```
count:          50
limit:          50
offset:         20
results:
  0:
    artifact_id:     "0-10_p021_bc77bf78"
    detail:          "statue"
    iconography:     "none"
    material:        "marble"
    page_num:        21
    pdf_name:        "0-10.pdf"
    period_precise:  "unknown"
    type:            "sculpture"
  1:
    artifact_id:     "0-10_p022_75d2072c"
    detail:          "unknown"
    iconography:     "none"
    material:        "unknown"
    page_num:        22
    pdf_name:        "0-10.pdf"
    period_precise:  "unknown"
    type:            "object"
  2:
    artifact_id:     "0-10_p023_9afad18b"
    detail:          "bead"
    iconography:     "none"
    material:        "gold"
    page_num:        23
    pdf_name:        "0-10.pdf"
    period_precise:  "5th century"
    type:            "glyptic"
```


http://127.0.0.1:5000/search_looted_artefacts?country=Netherlands


```
count:          10
country:        "Netherlands"
limit:          10
main_website:   null
offset:         0
results:
  0:
    artifact_id:     "0-25_p027_b022fb35"
    country:         "Netherlands"
    link:            "https://www.lot-art.com/auction-lots/CARVED-STONE-IDOLFIGURE-OF-PRE-ISLAMIC-SOUTH-ARABIAN-TYPE-possibly-Yemen/31-carved"
    normalized_domain: "lot-art.com"
  1:
    artifact_id:     "0-25_p036_bea25d9f"
    country:         "Netherlands"
    link:            "https://www.lot-art.com/auction-lots/Two-South-Arabian-Antiquities-Qataban-3rd-Century-BC1st-Century-AD/109-two_south-0
    normalized_domain: "lot-art.com"
  2:
    artifact_id:     "0-25_p037_76bcbec5"
    country:         "Netherlands"
    link:            "https://www.lot-art.com/auction-lots/Fine-South-Arabian-Alabaster-Male-Head/48-fine_south-14.7.22-artemi"
    normalized_domain: "lot-art.com"
  3:
    artifact_id:     "0-25_p038_a303509a"
    country:         "Netherlands"
    link:            "https://www.lot-art.com/auction-lots/Western-Asiatic-South-Arabian-Painted-Jar/1688-western_asiatic-04.6.20-timeline"
    normalized_domain: "lot-art.com"
  4:
    artifact_id:     "0-25_p039_21a0fd74"
    country:         "Netherlands"
    link:            "https://www.lot-art.com/auction-lots/Fine-South-Arabian-Qatabanian-Stone-Head/58-fine_south-10.9.20-artemi"
    normalized_domain: "lot-art.com"
  5:
```

GDPR (and more on data restrictions)

General Data Protection Regulation

This project does not involve any personal data. All collected and generated data relate exclusively to cultural artefacts, institutions, and publicly available sources, and therefore do not fall under GDPR constraints.

Restricted sensitive datasets

It is also important to note that, during the data collection phase, most existing datasets related to looted antiquities are either not publicly accessible or only available under strict conditions, including used datasets like mola artifacts.
For example:

- Interpol Stolen Works of Art database: https://www.interpol.int/Crimes/Cultural-heritage-crime/Stolen-Works-of-Art
- FBI National Stolen Art File: https://artcrimes.fbi.gov/

Access is limited due to the sensitive nature of the data (trafficking networks, security concerns) and requires specific authorisation rather than simple API access (even with a token). This reinforces the importance of structuring accessible sources.

Image restrictions

The extracted images used in this project were not stored in Git repository. Their ownership is not always clearly defined, as they originate either from the official Yemeni authority website or from external art market platforms. To avoid potential copyright issues and respect intellectual property, only references (paths or links) to the images were retained, rather than the files themselves.

Conclusions

Data pipeline / Vector comparison / Looted artifacts trafficking patterns

The POC (proof of concept) aimed to demonstrate the feasibility of implementing a complete pipeline, from PDF data extraction to its structuring in a DBMS and its exposure via an API. The results obtained confirm the viability of this approach, particularly yielding to usable and interoperable data. Furthermore, the comparison of images and text for each artefact, based on data vectorization, also proved successful, paving the way for relevant automated analyses.

The high similarity observed in both text and image scores led us to quickly implement a random sampling approach to compare cases with the smallest and largest distances (i.e., best and worst matches).
For text, some scores close to 0 still correspond to relatively distant descriptions. This suggests that while general text vectorization model is useful for capturing semantic proximity, it remains insufficient on their own for our specific needs. Indeed, the artefacts tend to share similar characteristics as they belong to the same cultural and heritage context, which limits the discriminative capacity of the model. For images, a similar approach showed that some pictures in the PDFs and on the websites were identical. We hypothesize that these images were reused across the web due to the lack of original photographs, especially for objects looted as early as 1994. At that time, access to cameras was limited and documentation relied mainly on physical storage, which may explain the reuse of available images.

Organizing the data into a relational database enabled the creation of queries leading to efficient data visualization, helping to identify initial patterns in the types of objects and their trajectories.
We can therefore conclude that the looted objects are:
- predominantly from Antiquity
- mainly objects, sculptures, jewelry, and currency featuring human and animal iconography
- the materials are not listed here because they are associated with the types of objects (for example, marble for sculpture, silver for currency)

Most Yermeni looted artifacts visible online are hosted on art dealers' websites based in the United States, UK, and Netherlands. These countries also rank among those that have returned the highest number of looted artifacts, according to MOLA's findings, suggesting a strong and consistent trajectory with the likely current location of these objects. In addition, the top three art dealers account for 54% of the recorded looted artifacts. It is important to consider potential biases behind this concentration, for instance by examining the data collection methodology before drawing firm conclusions. However, combined with the broader geographic trends observed, this pattern suggests that a closer focus on these key destinations could be particularly insightful.

While we have demonstrated that it is feasible to build a data pipeline and analyze the data, we have also encountered several challenges that need to be addressed, leading to suggestions for further improvement.

Main challenges

Extracting data from PDFs, combined with their translation, proved to be a particularly complex step. The organization of the information varied greatly from one document to another, revealing the absence of a standardized frame (tables, text blocks, dispatched images). This heterogeneity was initially addressed using PDF libraries that allowed content to be “chunked” based on entry points (table, picture) and relative positions on the page, but this approach had to be abandoned after several unsuccessful attempts. The use of an LLM ultimately proved more effective, allowing the information to be structured in JSON format, which formed the basis of the looted artifacts skeleton. Translation also remains a work in progress, particularly for specialized vocabulary, but also for certain languages that are less well-represented in open-source models, such as Arabic. These limitations result in considerable data cleaning work, particularly due to differences in encoding formats.

Further technical improvements

Considering these initial results, several avenues for improvement emerge. In particular, it would be worthwhile to continue developing structured databases in fields such as archaeology, to integrate more suitable embedding models and sublayers (such as Embedding Instructors) to improve text comparison performance, and to develop specialized LLMs that would be better suited to include linguistic diversity.

Github repository

https://github.com/anneleschallierdelisle/final_project_looted_artefacts